

Dual-View Hypercube Alignment Framework for Open-World Entity Typing

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Abstract

Traditional fine-grained entity typing (FET) operates under the closed-world assumption, where all classified types are limited within a predefined type ontology. However, as the world evolves, new entities of *unknown* types inevitably emerge in realistic open-world scenarios, falling outside the scope of the existing type ontology. To accommodate the open-world requirement, we investigate a new and critical task: open-world entity typing (OWET), which aims to simultaneously classify entity mentions within the known type ontology and discover mention clusters of unknown types. To tackle this task, we propose a Dual-View Hypercube Alignment (DVHA) framework, which leverages diverse tailor-designed alignment mechanisms based on hypercubes to jointly capture and combine complementary knowledge involved in the ontology and instance views for high-quality mention representation learning. In the ontology view, we propose exploiting the hypercube to represent the entity type and devise an intra-view hypercube alignment mechanism to refine optimal type hypercubes with effective exploitation of structural knowledge inherent in the type ontology. Whilst in the instance view, we design an innovative iterative point-hypercube alignment mechanism to learn discriminative mention representations by sufficiently harnessing the abundant semantic knowledge embodied in mentions and their textual contexts. To further consolidate the complementary knowledge delivered from both views, a cross-view partial alignment mechanism is developed by firstly establishing a cross-view bridge and then performing cross-view hypercube alignment between the two views, thereby facilitating more comprehensive and robust mention representations. Extensive experiments on two newly established OWET benchmark datasets demonstrate that our proposed DVHA framework consistently surpasses all the ten baselines by a large margin.

CCS Concepts

• Information systems → Information systems applications; Data mining.

Keywords

Fine-Grained Entity Typing; Open-World Entity Typing; Hypercube

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WWW '26, Dubai, United Arab Emirates
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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792694>

ACM Reference Format:

Hu Chen, Yu Zhu, and Wei Shen. 2026. Dual-View Hypercube Alignment Framework for Open-World Entity Typing. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792694>

1 Introduction

Fine-grained entity typing (FET), which aims to assign fine-grained types from a predefined type ontology to entity mentions based on their textual contexts, has made impressive contributions to the development of various knowledge-driven applications, including knowledge base population [17, 24], entity linking [6, 29], question answering [11, 43] and relation extraction [32, 42]. Traditional FET [8, 10, 25, 28, 31, 41] adheres to the *closed-world assumption*, i.e., all classified types belong to the type ontology provided by a specific knowledge base (KB).

Nevertheless, KBs are typically curated manually and inherently incomplete [27], encompassing only a limited number of entity types. Additionally, in realistic open-world scenarios, new entities of *unknown* types may frequently emerge, which do not fit into the existing type ontology. To accommodate the open-world requirement, we propose to explore a novel and crucial task, *open-world entity typing* (OWET), which involves a labeled training set exclusively comprising entity mentions of known types, and an unlabeled (test) set that contains mentions from both known and unknown types. The goal of the OWET task is to simultaneously classify entity mentions within the known type ontology and discover mention clusters of unknown types from the unlabeled set.

Under the OWET setting, the absence of supervised signals for unknown types causes the learning process to be dominated by labeled known-type entity mentions. This phenomenon inevitably introduces bias into the learned mention representations, making the model prone to misclassifying unknown-type mentions as known types, ultimately constraining its overall performance. Consequently, besides the utilization of the labeled set, a key to addressing the OWET task lies in sufficiently exploiting internal knowledge embedded within the unlabeled set to learn discriminative mention representations, thus promoting the model's generalization ability.

Meanwhile, the ontology provided in the OWET task exhibits a hierarchical structure composed of entity types, denoted as the *ontology view*, while entity mentions could be regarded as concrete instances instantiated from this ontology, constituting the *instance view*. Notably, the hierarchical structure within the ontology view provides valuable structural knowledge, whilst the textual context surrounding each entity mention in the instance view embodies abundant semantic knowledge. Accordingly, effectively integrating the complementary knowledge from both the ontology and instance views is expected to facilitate more comprehensive and

robust mention representation learning, thereby enhancing the model's discriminative capacity. However, the ontology view is characterized by a hierarchical structure with types connected via *is-a* relations, whereas the instance view is relatively flat, with each mention appearing independently. As a result, this intrinsic heterogeneity between the two views poses an evident obstacle for the effective fusion of their reciprocal knowledge.

To tackle the aforementioned issues, we propose a Dual-View Hypercube Alignment (DVHA) framework, which leverages varied alignment mechanisms on the basis of hypercubes to jointly capture and integrate complementary knowledge embedded in the ontology and instance views for superior mention representation learning.

Specifically, in the ontology view, conventional vector representations struggle to sufficiently capture the relationships between arbitrary entity types, i.e., inclusion and exclusion. To overcome this limitation, we propose employing the *hypercube* [37] to represent each entity type, termed as *type hypercube*. Compared with the traditional vector representation, hypercubes possess inherent geometric properties that are well-suited for modeling hierarchical relationships in the ontology view. In this way, as shown in Figure 1 (b) and 1 (c), the inclusion and exclusion between hypercubes can naturally reflect inclusion and exclusion relationships between entity types, substantiating it as a desirable representation choice. Ultimately, in order to construct optimal type hypercubes, we propose an *intra-view hypercube alignment* mechanism by unifying geometric and probabilistic constraints [18] to guarantee hierarchical relationships between type hypercubes.

Whilst in the instance view, to yield a strong initialization for mention representations, we apply supervised contrastive learning on the labeled set, alongside self-supervised learning on both the labeled and unlabeled sets. To further harness the knowledge embodied in the unlabeled set, we introduce an *iterative point-hypercube alignment* mechanism by refining the mention representations in an iterative manner. To be specific, it alternately performs clustering over mention representations and point-hypercube alignment to encourage each mention representation to align closely with its assigned cluster representation while remaining distant from others. Regarding the cluster representation, most existing methods [3, 4, 22] typically encode each cluster as a single-point prototype, failing to capture the diverse semantic knowledge encoded in multiple mentions within the cluster. To handle this issue, we represent each cluster as a more expressive hypercube, which constitutes a subspace encapsulating innumerable points while maintaining the cross-view geometric consistency with the type hypercube in the ontology view. Specifically, we develop an attentive fusion projection to aggregate mention representations of a cluster to derive its cluster hypercube, enabling the preservation of intricate semantic knowledge within the cluster. To align each mention representation with its corresponding cluster hypercube, we design a point-hypercube alignment strategy that locates single-point mention representations inside their corresponding cluster hypercubes, jointly optimizing mention representations and cluster hypercubes.

To incorporate the complementary knowledge delivered from both views for better representation modeling, we propose a *cross-view partial alignment* mechanism by firstly establishing a cross-view bridge and then performing cross-view hypercube alignment

between the two views. Notably, clusters formed within the instance view exhibit intrinsic semantic associations with specific mention types, including both known types existing in the ontology and unknown types outside it. Accordingly, we can establish a cross-view bridge based on the semantic association between cluster hypercubes (projected from instance-view clusters) and type hypercubes (derived from ontology-view types). However, two core challenges arise in the process of establishing this bridge between the instance view and the ontology view. Firstly, due to the incompleteness of the ontology which only covers known types, the cross-view type discrepancy naturally reveals an optimal matching problem between partial cluster hypercubes and type hypercubes. Secondly, only a limited portion of mentions within certain clusters are labeled, restricting supervised signals for guiding the matching process. To address these challenges, we employ integer linear programming (ILP) to establish the cross-view bridge, which optimally matches partial cluster hypercubes to type hypercubes via maximizing the matching accuracy based on the limited labeled set. Building upon the matching result, we develop a cross-view hypercube alignment strategy by jointly enforcing geometric and probabilistic constraints to align matched cluster and type hypercubes, thereby integrating the synergistic knowledge of the two views into the mention representations.

Our major contributions can be summarized as follows:

- To the best of our knowledge, we make the first attempt to explore the open-world entity typing (OWET) task, a novel and increasingly pivotal problem due to its practical applications.
- To tackle this novel OWET task, we develop a Dual-View Hypercube Alignment (DVHA) framework, which utilizes various tailor-made alignment mechanisms based on hypercubes to capture and combine reciprocal knowledge involved in the ontology view and the instance view.
- To effectively capitalize on the complementary knowledge delivered from both views for high-quality representation learning, we devise a cross-view partial alignment mechanism that firstly establishes a cross-view bridge and then conducts cross-view hypercube alignment between the two views.
- We create two benchmark datasets tailored for this new OWET task. Experiments on these datasets demonstrate that our DVHA framework consistently exceeds all the ten baselines.

2 Problem Formulation

Given a predefined type ontology $\mathcal{O}$ and a labeled training set $\mathcal{M}_l$, the fine-grained entity typing (FET) task aims to assign each testing entity mention m an appropriate type t based on its context c . Formally, the type ontology $\mathcal{O} = \{t_k\}_{k=1}^{|\mathcal{O}|}$ typically exhibits a hierarchical structure composed of multi-granularity types, and the labeled training set $\mathcal{M}_l = \{(m_i, c_i)\}_{i=1}^{|\mathcal{M}_l|}$ encompasses $|\mathcal{M}_l|$ entity mentions along with their corresponding contexts, where each entity mention m_i within its context c_i is labeled with a known type $\hat{t}_i \in \mathcal{O}$. For the unlabeled (test) set $\mathcal{M}_u = \{(m_i, c_i)\}_{i=1}^{|\mathcal{M}_u|}$, the traditional FET task under the closed-world assumption necessitates that all entity mentions in the unlabeled set $\mathcal{M}_u$ are classified into known types within the predefined type ontology $\mathcal{O}$. While in a more realistic open-world scenario, some entity mentions within the unlabeled set $\mathcal{M}_u$ may come from unknown types that do not belong to the predefined type ontology $\mathcal{O}$. Consequently, the

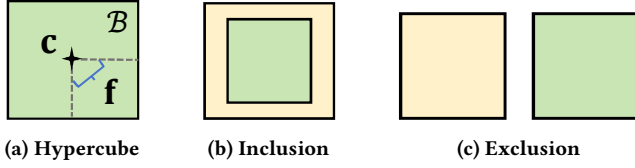


Figure 1: Illustration of hypercubes.

ultimate goal of the OWET task is to simultaneously classify entity mentions of known types within the type ontology $\mathcal{O}$ and discover unknown-type mention clusters from the unlabeled set $\mathcal{M}_u$.

3 The Proposed Framework: DVHA

In this section, we present a detailed description of our proposed framework DVHA, as depicted in Figure 2. We begin with the modeling of the instance view (Sec. 3.1) and the ontology view (Sec. 3.2) to capture their distinct embodied knowledge. Afterwards, a cross-view partial alignment mechanism (Sec. 3.3) is designed to effectively integrate the complementary knowledge from both views, thereby promoting superior representation learning.

3.1 Instance-View Modeling

To capture the abundant semantic knowledge within the instance view for mention representation learning, we first initialize the mention representations (Sec. 3.1.1) and subsequently devise an iterative point-hypercube alignment mechanism (Sec. 3.1.2) to facilitate discriminative mention representations.

3.1.1 Mention Representation Initialization. In the FET task, a widely used paradigm [8, 25, 31, 41] is to leverage pre-trained language models (PLMs) to obtain mention representations for entity typing. Given an entity mention m_i and its surrounding context c_i from both labeled set $\mathcal{M}_l$ and unlabeled set $\mathcal{M}_u$, we first construct its input sequence by concatenating them with two special tokens, [CLS] and [SEP]. Then, a PLM (e.g., BERT [9]) is employed as the text encoder to derive the representation of the input sequence, formulated as follows:

$$\mathbf{h}_i = \text{PLM}([\text{CLS}]m_i[\text{SEP}]c_i[\text{SEP}]). \quad (1)$$

Here, the [CLS] token representation $\mathbf{h}_i \in \mathbb{R}^d$ is regarded as the representation of the mention m_i within the context c_i .

In order to yield a strong mention representation initialization (① in Figure 2), we jointly apply a supervised contrastive loss $\mathcal{L}_{sc}$ [20] on the labeled set and a masked language modeling loss $\mathcal{L}_{mlm}$ [9] on both labeled and unlabeled sets:

$$\mathcal{L}_{init} = \mathcal{L}_{sc}(\mathcal{M}_l) + \mathcal{L}_{mlm}(\mathcal{M}_l \cup \mathcal{M}_u). \quad (2)$$

Here, the supervised contrastive loss exploits the known-type supervised signals embodied within the labeled set by pulling mentions of the same type closer and pushing those of different types farther apart, while the masked language modeling loss encourages the model to predict randomly masked tokens within each input sequence, thus capturing its rich contextual semantics.

3.1.2 Iterative Point-Hypercube Alignment. Due to the absence of supervised signals for unknown types, the mention representation initialization is dominated by labeled known-type mentions, inevitably resulting in biased mention representations, ultimately making the model susceptible to misclassifying unknown-type mentions as known types. To mitigate this issue, we design an iterative point-hypercube alignment mechanism (② in Figure 2) by performing clustering over mention representations and point-hypercube alignment alternately. This iterative procedure enables

the refinement of mention representations by sufficiently harnessing the internal knowledge within the unlabeled set, enhancing their discriminative power.

Mention Cluster Generation. In order to facilitate discriminative mention representations, it is crucial to establish distinct boundaries between entity mentions of different types, while mentions of the same type should be close together in the representation space. As we know, clustering is a simple yet effective way for organizing semantics and analyzing data distribution [5, 40], and thus can be incorporated to guide this process of mention representation refinement. However, estimating the appropriate granularity of clusters remains a non-trivial challenge for the clustering process. As outlined in Sec. 2, the labeled set contains type labels of a specific granularity for entity mentions, whereas the unlabeled set lacks such label information. To deal with this challenge, we adopt hierarchical agglomerative clustering (③ in Figure 2) over mention representations from both labeled and unlabeled sets by sufficiently capitalizing on the label information embodied in the labeled set to estimate the optimal clustering granularity. Concretely, during clustering, we progressively merge the closest pair of mention clusters until maximizing the clustering accuracy over the labeled set, ultimately partitioning mentions from both labeled and unlabeled sets into clusters $\mathcal{U} = \{\mathcal{U}_v\}_{v=1}^{|\mathcal{U}|}$ with optimal granularity.

Point-Hypercube Alignment. Based on the above clustering result, we design a point-hypercube alignment strategy (④ in Figure 2) to drive each mention representation to align closely with its assigned cluster representation while remaining distant from others, thereby facilitating more discriminative mention representations.

• **Cluster Hypercube Projection.** With respect to the cluster representation, most existing methods [3, 4, 22] typically encode each cluster as a single-point prototype, which inherently struggles to capture the diverse semantic knowledge encoded in multiple mentions within the cluster. In light of this limitation, we propose employing the *hypercube* [37] to represent each cluster, defined as *cluster hypercube*, which offers a more expressive cluster representation.

Geometrically, a hypercube is a d -dimensional analogue of a cube. For the convenience of computation, a hypercube can be characterized by a center and an offset, both residing in the same d dimensions, as shown in Figure 1 (a). Formally, in a d -dimensional space, a hypercube $\mathcal{B} = (\mathbf{c}, \mathbf{f}) \in \mathbb{R}^{2d}$ can be defined as:

$$\mathcal{B} \equiv \{\mathbf{b} \in \mathbb{R}^d : \mathbf{c} - \mathbf{f} \preceq \mathbf{b} \preceq \mathbf{c} + \mathbf{f}\}, \quad (3)$$

where $\preceq$ denotes element-wise inequality and $\mathbf{b}$ represents any point contained within the hypercube $\mathcal{B}$. Here, $\mathbf{c} \in \mathbb{R}^d$ is the center of the hypercube and $\mathbf{f} \in \mathbb{R}^d$ represents its offset. According to the definition, a hypercube spans a d -dimensional subspace encompassing innumerable points, empowering it to preserve intricate semantic knowledge within the cluster.

To project a cluster into its corresponding cluster hypercube, an intuitive approach is to determine the hypercube boundary by computing the minimum and maximum corners over all mention representations of this cluster. However, this method treats all mentions equally without accounting for their relative importance, which renders it sensitive to outliers and may lead to excessively large hypercubes. To alleviate this issue, we devise an attentive fusion projection to aggregate mention representations of each cluster $\mathcal{U}_v$ to derive its cluster hypercube $\mathcal{B}_v^T = (\mathbf{c}_v^T, \mathbf{f}_v^T)$. Specifically, the

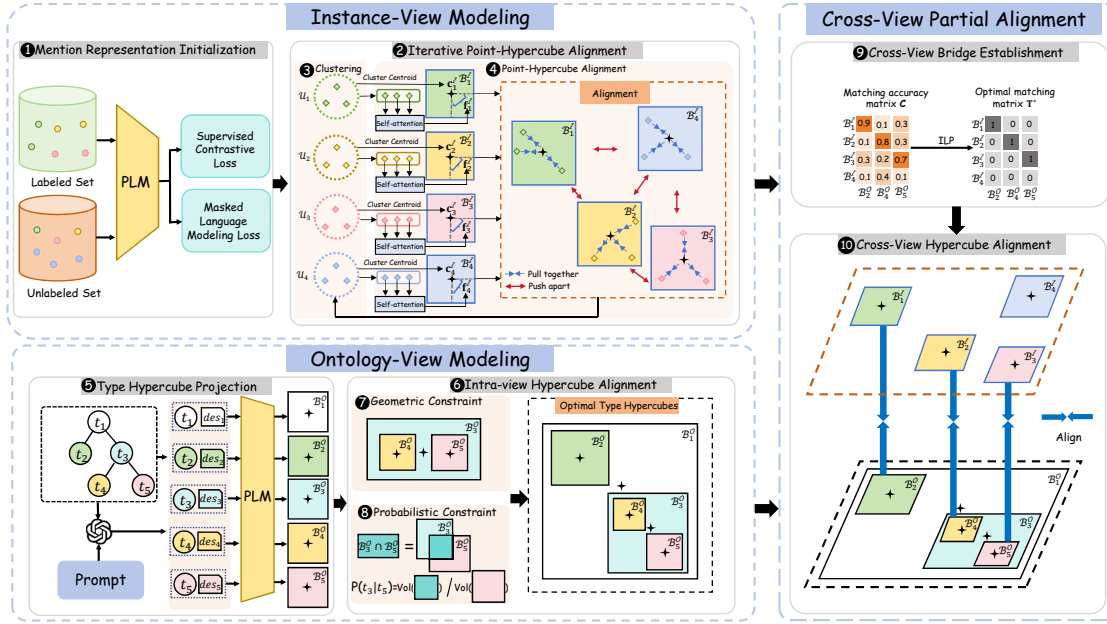


Figure 2: The overall architecture of DVHA.

cluster centroid of the cluster $\mathcal{U}_v$ is adopted as the center $\mathbf{c}_v^I$ of the hypercube $\mathcal{B}_v^I$ to capture the core semantics of the cluster $\mathcal{U}_v$. Then, we employ a self-attention mechanism [36] by accounting for each mention's importance discrepancy within the cluster $\mathcal{U}_v$ to learn the offset $\mathbf{f}_v^I$ of the hypercube $\mathcal{B}_v^I$, formulated as:

$$\mathbf{c}_v^I = \frac{1}{|\mathcal{U}_v|} \sum_{m_i \in \mathcal{U}_v} \mathbf{h}_i, \quad \mathbf{f}_v^I = \text{relu}(\mathbf{W}^V \mathbf{H}_v \text{softmax}(\frac{\mathbf{W}^E \mathbf{H}_v \mathbf{q}}{\sqrt{d}})), \quad (4)$$

where $\mathbf{q}$ is the learnable query vector, $\mathbf{W}^E, \mathbf{W}^V \in \mathbb{R}^{d \times d}$ denote the learnable key and value projection matrices respectively, and $\mathbf{H}_v$ is the matrix stacking all mention representations of cluster $\mathcal{U}_v$. The relu activation function ensures the non-negativity of the offset. In summary, this attentive fusion allows us to project clusters $\mathcal{U} = \{\mathcal{U}_v\}_{v=1}^{|\mathcal{U}|}$ into cluster hypercubes $\mathcal{B}^I = \{\mathcal{B}_v^I\}_{v=1}^{|\mathcal{U}|}$ in a more expressive geometric manner.

• **Alignment.** After constructing the cluster hypercubes, we subsequently align mention representations with their corresponding cluster hypercubes by locating each single-point mention representation inside its corresponding cluster hypercube, thus jointly optimizing mention representations and cluster hypercubes.

Before alignment, following [1], we define the point-hypercube distance function $\text{Dist}(\mathbf{h}_i, \mathcal{B}_v^I)$ to quantify the distance between a mention representation $\mathbf{h}_i$ and a cluster hypercube $\mathcal{B}_v^I = (\mathbf{c}_v^I, \mathbf{f}_v^I)$:

$$\text{Dist}(\mathbf{h}_i, \mathcal{B}_v^I) = \begin{cases} \|(\mathbf{h}_i - \mathbf{c}_v^I) / (2\mathbf{f}_v^I + 1)\|_2 & \text{if } \mathbf{h}_i \in \mathcal{B}_v^I, \\ \|(\mathbf{h}_i - \mathbf{c}_v^I) \circ (2\mathbf{f}_v^I + 1) - \kappa\|_2 & \text{otherwise,} \end{cases} \quad (5)$$

where $\kappa = \mathbf{f}_v^I \circ (2\mathbf{f}_v^I - 1 / (2\mathbf{f}_v^I + 1) + 1)$, is an offset-dependent scaling factor, and $\circ$ denotes the element-wise multiplication operation. This function yields a slow-growing distance for mention representations inside the hypercube, but heavily penalizes those outside

it, therefore driving mention representations more effectively into their assigned cluster hypercubes.

Based on this point-hypercube distance, we introduce a point-hypercube alignment loss $\mathcal{L}_{p-h}$ to enforce each mention representation $\mathbf{h}_i$ to lie inside its designated cluster hypercube $\mathcal{B}_{m_i}^I$ while pushing it away from all negative cluster hypercubes, defined as:

$$\mathcal{L}_{p-h} = -\text{logsigmoid}(\gamma - \text{Dist}(\mathbf{h}_i, \mathcal{B}_{m_i}^I)) - \sum_{\tilde{\mathcal{B}}_{m_i}^I \in \mathcal{B}^I \setminus \mathcal{B}_{m_i}^I} w(\tilde{\mathcal{B}}_{m_i}^I | \mathbf{h}_i) \text{logsigmoid}(\text{Dist}(\mathbf{h}_i, \tilde{\mathcal{B}}_{m_i}^I) - \gamma), \quad (6)$$

where γ is a margin hyperparameter, $\tilde{\mathcal{B}}_{m_i}^I$ indicates a negative cluster hypercube and $w(\tilde{\mathcal{B}}_{m_i}^I | \mathbf{h}_i)$ is regarded as its importance weight w.r.t. $\mathbf{h}_i$, defined as:

$$w(\tilde{\mathcal{B}}_{m_i}^I | \mathbf{h}_i) = \frac{\exp(-\tau \text{Dist}(\mathbf{h}_i, \tilde{\mathcal{B}}_{m_i}^I))}{\sum_{\tilde{\mathcal{B}}_r^I \in \mathcal{B}^I \setminus \mathcal{B}_{m_i}^I} \exp(-\tau \text{Dist}(\mathbf{h}_i, \tilde{\mathcal{B}}_r^I))}, \quad (7)$$

where τ is a temperature parameter. This weight function emphasizes hard negative cluster hypercubes by assigning them higher importance weights during alignment. To prevent the learned cluster hypercubes from collapsing to overly small volumes, we implement an additional regularization loss $\mathcal{L}_{reg}^I$ to enforce a minimum offset size ϕ^I for each cluster hypercube $\mathcal{B}_v^I = (\mathbf{c}_v^I, \mathbf{f}_v^I)$:

$$\mathcal{L}_{reg}^I = \frac{1}{d} \sum_{i=1}^d \min(0, \mathbf{f}_{v,i}^I - \phi^I)^2, \quad (8)$$

where i indexes the representation dimension. To avoid catastrophic forgetting of supervisory knowledge involved in the labeled set, we additionally incorporate the supervised contrastive loss $\mathcal{L}_{sc}$ (in Eq. 2) during alignment. Eventually, the overall loss $\mathcal{L}_{align}^I$ for the point-hypercube alignment strategy is formulated as:

$$\mathcal{L}_{align}^I = \mathcal{L}_{p-h} + \mathcal{L}_{reg}^I + \mathcal{L}_{sc}. \quad (9)$$

3.2 Ontology-View Modeling

In addition to the extensive semantic knowledge encoded in the instance view, the hierarchical structure within the ontology view provides valuable structural knowledge. Consequently, we project each entity type in the ontology view into a type hypercube (Sec. 3.2.1) and develop an intra-view hypercube alignment mechanism (Sec. 3.2.2) to learn optimal type hypercubes with effective exploitation of structural knowledge embodied in the ontology view.

3.2.1 Type Hypercube Projection. It is worth noting that the ontology view in the OWET task exhibits a hierarchical structure composed of entity types. As a consequence, conventional vector representations unavoidably encounter inherent limitations in characterizing relations between arbitrary entity types, i.e., inclusion and exclusion. To overcome this deficiency, we propose projecting each entity type into a hypercube (5 in Figure 2), termed as *type hypercube*. Compared with traditional vector representations, the geometric properties of hypercubes intrinsically support the modeling of hierarchical relationships in the ontology view. In this way, as illustrated in Figure 1 (b) and 1 (c), the inclusion and exclusion relationships between hypercubes can naturally reflect inclusion and exclusion relationships between entity types, demonstrating its suitability as a desirable representation choice. Nevertheless, projecting type hypercubes solely based on entity type names often fails to precisely interpret and differentiate these entity types. Therefore, to provide comprehensive semantic contexts for the type hypercube projection, we incorporate a tailor-designed prompt (as shown in Figure 4) to instruct large language models (LLMs) in generating an elaborate description for each entity type. Each type's generated description is prompted to maintain semantic relevance to its parent type while remaining distinct from its sibling types, thereby supplying comprehensive and distinguishable semantic contexts for projection.

In the projection process, given a type t_k with its generated description des_k , we first concatenate them with special tokens and encode the concatenated sequence through a PLM to obtain the [CLS] token representation $\mathbf{h}_{[CLS]} \in \mathbb{R}^d$. This representation is then transformed by a highway layer $\mathcal{F}_{proj}$ into a $2d$ -dimensional representation, which is subsequently split into two equal-length components. These components are interpreted as the center $\mathbf{c}_k^O$ and offset $\mathbf{f}_k^O$ of the projected type hypercube $\mathcal{B}_k^O$:

$$\mathcal{B}_k^O = (\mathbf{c}_k^O, \mathbf{f}_k^O) = \mathcal{F}_{proj}(\text{PLM}([\text{CLS}]t_k[\text{SEP}]des_k[\text{SEP}])). \quad (10)$$

Note that the offset $\mathbf{f}_k^O$ is applied with relu activation function to ensure its non-negativity. Finally, we project all types $\{t_k\}_{k=1}^{|O|}$ in the type ontology O into type hypercubes $\mathcal{B}^O = \{\mathcal{B}_k^O\}_{k=1}^{|O|}$, thereby yielding a desirable type representation in the ontology view.

3.2.2 Intra-view Hypercube Alignment. To learn optimal type hypercubes, we introduce an intra-view hypercube alignment mechanism (6 in Figure 2) that incorporates both geometric and probabilistic constraints [18] to guarantee the inclusion and exclusion relationships between type hypercubes.

Geometric Constraint. Specifically, for each entity type t_k within the ontology, we regard its parent type t_p as a positive type t_{k+} , thus capturing the inclusion relationship between t_k and t_{k+} . To reflect the inclusion relation between corresponding type hypercubes from the geometric perspective (7 in Figure 2), the type hypercube $\mathcal{B}_k^O$

(projected from t_k) is required to be entirely enclosed within the positive type hypercube $\mathcal{B}_{k+}^O$ (projected from t_{k+}). Formally, the minimum and maximum corners of $\mathcal{B}_k^O$ are computed as $\mathbf{l}_k^O = \mathbf{c}_k^O - \mathbf{f}_k^O$ and $\mathbf{r}_k^O = \mathbf{c}_k^O + \mathbf{f}_k^O$, respectively. Similarly, the minimum and maximum corners of $\mathcal{B}_{k+}^O$ could be computed in a similar way and are denoted as $\mathbf{l}_{k+}^O$ and $\mathbf{r}_{k+}^O$, respectively. The inclusion relation between $\mathcal{B}_k^O$ and $\mathcal{B}_{k+}^O$ is expressed as:

$$\mathbf{l}_{k,i}^O \geq \mathbf{l}_{k+,i}^O, \quad \mathbf{r}_{k,i}^O \leq \mathbf{r}_{k+,i}^O, \quad \forall i \in \{1, 2, \dots, d\}, \quad (11)$$

where i indexes the representation dimension. Accordingly, we derive the positive geometric constraint loss $\mathcal{L}_{geo}^+$, formalized as:

$$\mathcal{L}_{geo}^+(\mathcal{B}_k^O, \mathcal{B}_{k+}^O, \delta) = \frac{1}{2d} \sum_{i=1}^d (([\mathbf{l}_{k+,i}^O - \mathbf{l}_{k,i}^O + \delta]_+)^2 + ([\mathbf{r}_{k,i}^O - \mathbf{r}_{k+,i}^O + \delta]_+)^2), \quad (12)$$

where $[\cdot]_+ = \max(0, \cdot)$ and δ denotes a positive geometric margin.

On the other hand, a negative type set $Q(t_k)$ is constructed from types that are not ancestors of t_k in the ontology O , thereby indicating an exclusion relation between t_k and each negative type $t_{k-} \in Q(t_k)$. To satisfy the exclusion relation between corresponding type hypercubes, the type hypercubes $\mathcal{B}_k^O$ must be disjoint from each negative type hypercube $\mathcal{B}_{k-}^O$ (projected from t_{k-}). Denoting the minimum and maximum corners of the hypercube $\mathcal{B}_{k-}^O$ as $\mathbf{l}_{k-}^O$ and $\mathbf{r}_{k-}^O$, the exclusion relation between $\mathcal{B}_k^O$ and $\mathcal{B}_{k-}^O$ requires:

$$\mathbf{l}_{k,max,i}^O \geq \mathbf{r}_{k,min,i}^O, \quad \forall i \in \{1, 2, \dots, d\}, \quad (13)$$

$$\mathbf{l}_{k,max,i}^O = \max(\mathbf{l}_{k,i}^O, \mathbf{l}_{k-,i}^O), \quad \mathbf{r}_{k,min,i}^O = \min(\mathbf{r}_{k,i}^O, \mathbf{r}_{k-,i}^O).$$

Hence, the negative geometric constraint loss $\mathcal{L}_{geo}^-$ is given by:

$$\mathcal{L}_{geo}^-(\mathcal{B}_k^O, \mathcal{B}_{k-}^O, \epsilon) = \frac{1}{d} \sum_{i=1}^d ([(\mathbf{r}_{k,min,i}^O - \mathbf{l}_{k,max,i}^O)/2 + \epsilon]_+)^2, \quad (14)$$

where ϵ is a negative geometric margin.

Probabilistic Constraint. From the probabilistic perspective (8 in Figure 2), the inclusion relation between t_k and t_{k+} can be redefined as a conditional probability $P(t_{k+}|t_k)=1$, whereas the exclusion relation between t_k and t_{k-} implies $P(t_{k-}|t_k)=0$. As a result, optimal type hypercubes should satisfy these conditional probabilities for both inclusion and exclusion relations. Inspired by prior hypercube-based studies [18, 28], the normalized volume of the intersection between type hypercubes $\mathcal{B}_x^O$ and $\mathcal{B}_y^O$ can be employed to evaluate the conditional probability $P(t_y|t_x)$ between types t_x and t_y , characterized as:

$$P(t_y|t_x) = P(\mathcal{B}_y^O|\mathcal{B}_x^O) = \frac{\text{Vol}(\mathcal{B}_y^O \cap \mathcal{B}_x^O)}{\text{Vol}(\mathcal{B}_x^O)}, \quad (15)$$

$$\text{Vol}(\mathcal{B}_x^O) = \prod_{i=1}^d (r_{x,i}^O - l_{x,i}^O),$$

$$\text{Vol}(\mathcal{B}_y^O \cap \mathcal{B}_x^O) = \prod_{i=1}^d [\min(r_{x,i}^O, r_{y,i}^O) - \max(l_{x,i}^O, l_{y,i}^O)]_+,$$

where the $\text{Vol}(\cdot)$ indicates the volume of a hypercube. On this basis, we develop a positive probabilistic constraint loss $\mathcal{L}_{pro}^+$ for type hypercubes $\mathcal{B}_k^O$ and $\mathcal{B}_{k+}^O$ as:

$$\mathcal{L}_{pro}^+(\mathcal{B}_k^O, \mathcal{B}_{k+}^O) = (1 - P(\mathcal{B}_{k+}^O|\mathcal{B}_k^O))^2, \quad (16)$$

and a negative probabilistic constraint loss $\mathcal{L}_{pro}^-$ for type hypercubes $\mathcal{B}_k^O$ and $\mathcal{B}_{k-}^O$ as:

$$\mathcal{L}_{pro}^-(\mathcal{B}_k^O, \mathcal{B}_{k-}^O) = (0 - P(\mathcal{B}_{k-}^O|\mathcal{B}_k^O))^2. \quad (17)$$

Moreover, similar to Eq. 8, we also introduce a regularization loss $\mathcal{L}_{reg}^O$ to guarantee a minimum offset size ϕ^O for each type hypercube $\mathcal{B}_k^O$:

$$\mathcal{L}_{reg}^O = \frac{1}{d} \sum_{i=1}^d \min(0, \mathbf{f}_{k,i}^O - \phi^O)^2. \quad (18)$$

Finally, to ensure the inclusion and exclusion relations between type hypercubes, we incorporate the aforementioned geometric constraint, probabilistic constraint and regularization losses in the intra-view hypercube alignment mechanism. The overall loss $\mathcal{L}_{align}^O$ is defined as:

$$\mathcal{L}_{align}^O = \mathcal{L}_{geo}^+ + \mathcal{L}_{geo}^- + \alpha(\mathcal{L}_{pro}^+ + \mathcal{L}_{pro}^-) + \mathcal{L}_{reg}^O, \quad (19)$$

where α controls the contribution of the probabilistic constraint.

3.3 Cross-View Partial Alignment

To effectively integrate the complementary knowledge delivered from both instance and ontology views, we propose a cross-view partial alignment mechanism that firstly establishes a cross-view bridge (Sec. 3.3.1) and then conducts cross-view hypercube alignment (Sec. 3.3.2) between the two views, thereby facilitating more comprehensive and robust mention representation learning.

3.3.1 Cross-View Bridge Establishment. Notably, clusters formed within the instance view inherently exhibit semantic association with specific mention types, including both known types within the ontology and unknown types outside it. Thus, we can establish a cross-view bridge (9 in Figure 2) based on the semantic association between cluster hypercubes (projected from instance-view clusters) and type hypercubes (derived from ontology-view types).

Nevertheless, two core challenges emerge in the process of establishing this bridge between the instance view and the ontology view. Firstly, due to the incompleteness of the ontology which only covers known types, the cross-view type discrepancy naturally reveals an optimal matching problem between partial cluster hypercubes and type hypercubes. Secondly, only a limited portion of mentions within certain clusters are labeled, constraining supervised signals for guiding the matching process. To tackle these challenges, we adopt integer linear programming (ILP) to establish a cross-view bridge, which optimally matches partial cluster hypercubes to type hypercubes via maximizing the matching accuracy based on the limited labeled set. Notably, the mentions in existing FET datasets [14, 38] are typically annotated with bottom-level types within the type ontology $\mathcal{O}$, which motivates us to establish the cross-view bridge between cluster hypercubes and type hypercubes projected from these bottom-level types $\dot{\mathcal{O}}$. Formally, we first define a binary matching matrix $\mathbf{T} \in \{0, 1\}^{|\mathcal{U}| \times |\dot{\mathcal{O}}|}$, where $T_{v,k}$ denotes whether the cluster hypercube $\mathcal{B}_v^I$ is matched to type hypercube $\mathcal{B}_k^O$. To sufficiently exploit the labeled mentions in establishing this cross-view bridge, we extract labeled mentions contained in each cluster $\mathcal{U}_v$ to form its subcluster $\dot{\mathcal{U}}_v$, which corresponds to cluster hypercube $\dot{\mathcal{B}}_v^I$. Meanwhile, we construct a subset $\mathcal{M}_I(t_k)$ of labeled mentions in $\mathcal{M}_I$ annotated with type t_k , which correlates with type hypercube $\mathcal{B}_k^O$. Then, we introduce a matching accuracy matrix $\mathbf{C}$, where $C_{v,k}$ represents the matching accuracy between $\mathcal{B}_v^I$ and $\mathcal{B}_k^O$ by calculating the Jaccard similarity between $\dot{\mathcal{U}}_v$ and $\mathcal{M}_I(t_k)$:

$$C_{v,k} = |\dot{\mathcal{U}}_v \cap \mathcal{M}_I(t_k)| / |\dot{\mathcal{U}}_v \cup \mathcal{M}_I(t_k)|. \quad (20)$$

Finally, to obtain the optimal matching result between partial cluster hypercubes and type hypercubes, we employ ILP to search for the optimal matching matrix $\mathbf{T}^*$ that maximizes the overall matching accuracy, formalized as:

$$\begin{aligned} \mathbf{T}^* = \arg\max_{\mathbf{T} \in \Delta} \langle \mathbf{C}, \mathbf{T} \rangle_F &= \sum_{v=1}^{|\mathcal{U}|} \sum_{k=1}^{|\dot{\mathcal{O}}|} C_{v,k} T_{v,k}, \\ \text{s.t. } \Delta = \{ \mathbf{T} \in \{0, 1\}^{|\mathcal{U}| \times |\dot{\mathcal{O}}|} \mid &\mathbf{T} \mathbf{1}_{|\dot{\mathcal{O}}|} \leq \mathbf{1}_{|\mathcal{U}|}, \mathbf{T}^\top \mathbf{1}_{|\mathcal{U}|} = \mathbf{1}_{|\dot{\mathcal{O}}|} \}, \end{aligned} \quad (21)$$

where $\langle \cdot, \cdot \rangle_F$ denotes the Frobenius inner product between two matrices and $\mathbf{1}_{|\mathcal{U}|}$ indicates an $|\mathcal{U}|$ -dimensional all-one vector. To satisfy that each cluster hypercube is matched at most once while each type hypercube is required to be matched exactly once, the summation of each row in $\mathbf{T}$ is bounded by 1, while the summation of each column is constrained to 1, as specified in Eq. 21. Based on the obtained optimal matching matrix $\mathbf{T}^*$, we derive the matching result, i.e., $|\dot{\mathcal{O}}|$ matched cluster hypercubes $\dot{\mathcal{B}}^I = \{\mathcal{B}_v^I\}_{v=1}^{|\dot{\mathcal{O}}|}$ corresponding to $|\dot{\mathcal{O}}|$ type hypercubes $\dot{\mathcal{B}}^O = \{\mathcal{B}_k^O\}_{k=1}^{|\dot{\mathcal{O}}|}$.

3.3.2 Cross-View Hypercube Alignment. On the basis of the matching result, we propose a cross-view hypercube alignment strategy (10 in Figure 2) that unifies geometric and probabilistic constraints to align matched cluster and type hypercubes, thereby integrating the synergistic knowledge of the two views into the mention representations. Nonetheless, the distinct modeling of the two heterogeneous views inherently leads to divergent representation spaces, which pose significant challenges for the cross-view hypercube alignment. To overcome this challenge, we first introduce a cross-view transformation function to transform each matched cluster hypercube into the space of type hypercubes, enabling cross-view hypercube alignment within a unified representation space. In specific, for each matched cluster hypercube $\mathcal{B}_v^I = (\mathbf{c}_v^I, \mathbf{f}_v^I)$, we transform it into $\dot{\mathcal{B}}_v^I = (\dot{\mathbf{c}}_v^I, \dot{\mathbf{f}}_v^I) = (\mathbf{W}_c \mathbf{c}_v^I, \text{relu}(\mathbf{W}_f \mathbf{f}_v^I))$, where $\mathbf{W}_c, \mathbf{W}_f \in \mathbb{R}^{d \times d}$ denote the learnable transformation matrices for the center and offset, respectively. Subsequently, to ensure effective alignment between matched cluster and type hypercubes, each matched cluster hypercube $\dot{\mathcal{B}}_v^I$ and its corresponding type hypercube $\mathcal{B}_k^O$ are required to satisfy mutual inclusion, while $\dot{\mathcal{B}}_v^I$ and $\mathcal{B}_k^O$ remain exclusion from each negative type hypercube $\mathcal{B}_{k^-}^O$ (constructed in the same way as described in Sec. 3.2.2).

Accordingly, following the previous intra-view hypercube alignment mechanism (Sec. 3.2.2), we design a cross-view geometric constraint loss $\mathcal{L}_{geo}^{Cross}$ to ensure the geometric constraint:

$$\begin{aligned} \mathcal{L}_{geo}^{Cross} &= \mathcal{L}_{geo}^+(\mathcal{B}_k^O, \dot{\mathcal{B}}_v^I, 0) + \mathcal{L}_{geo}^+(\dot{\mathcal{B}}_v^I, \mathcal{B}_k^O, 0) + \\ &\quad \mathcal{L}_{geo}^-(\mathcal{B}_k^O, \mathcal{B}_{k^-}^O, \epsilon) + \mathcal{L}_{geo}^-(\dot{\mathcal{B}}_v^I, \mathcal{B}_{k^-}^O, \epsilon). \end{aligned} \quad (22)$$

Here, we set the positive geometric margin $\delta = 0$ in $\mathcal{L}_{geo}^+$ to enforce the mutual inclusion between matched cluster and type hypercubes, and ϵ is a negative geometric margin (the same as in Eq. 14). Simultaneously, the cross-view probabilistic constraint loss $\mathcal{L}_{pro}^{Cross}$ for the probabilistic constraint is defined as:

$$\begin{aligned} \mathcal{L}_{pro}^{Cross} &= \mathcal{L}_{pro}^+(\mathcal{B}_k^O, \dot{\mathcal{B}}_v^I) + \mathcal{L}_{pro}^+(\dot{\mathcal{B}}_v^I, \mathcal{B}_k^O) + \\ &\quad \mathcal{L}_{pro}^-(\mathcal{B}_k^O, \mathcal{B}_{k^-}^O) + \mathcal{L}_{pro}^-(\dot{\mathcal{B}}_v^I, \mathcal{B}_{k^-}^O). \end{aligned} \quad (23)$$

Combining the geometric and probabilistic constraints, the overall loss for the cross-view hypercube alignment strategy is defined as:

$$\mathcal{L}_{align}^{Cross} = \mathcal{L}_{geo}^{Cross} + \alpha \mathcal{L}_{pro}^{Cross}, \quad (24)$$

Table 1: Results on BBN and OntoNotes. The best result in each metric is in boldface and the second-best result is underlined.

Methods	BBN			OntoNotes		
	Acc-known	Acc-unknown	Acc-all	Acc-known	Acc-unknown	Acc-all
CDAC+ [23]	0.109	0.127	0.113	0.120	0.198	0.143
DAC [44]	0.147	0.156	0.149	0.116	0.142	0.123
MTP [46]	0.295	0.266	0.289	0.241	0.313	0.261
USNID [45]	0.501	0.339	0.464	0.369	0.278	0.343
DWGF [35]	0.276	0.304	0.283	0.312	0.321	0.314
TAN [3]	0.646	0.259	0.558	0.436	0.219	0.374
KTN [34]	0.386	0.477	0.406	0.244	0.362	0.277
SDC [2]	0.318	0.442	0.346	0.208	0.313	0.238
GPT-4.1 [30]	0.591	0.310	0.527	0.256	0.556	0.343
DeepSeek-R1 [15]	0.702	0.658	0.692	0.463	0.570	0.494
DVHA (Ours)	0.911	0.807	0.887	0.900	0.589	0.808

where α indicates a control parameter (the same as in Eq. 19).

3.4 Overall Training

To jointly capture and combine complementary knowledge embedded in the instance and ontology views, we propose an overall training procedure for our DVHA framework. In the instance view, we perform iterative point-hypercube alignment mechanism (Sec. 3.1.2) based on the initialized mention representations (Sec. 3.1.1), thereby refining mention representations and cluster hypercubes progressively. Whilst in the ontology view, we project each entity type into a type hypercube (Sec. 3.2.1) and optimize these type hypercubes through intra-view hypercube alignment mechanism (Sec. 3.2.2). Building on these obtained cluster and type hypercubes at each iteration, we perform cross-view partial alignment mechanism (Sec. 3.3) to effectively integrate the complementary knowledge delivered from both views. Consequently, the overall training objective at the e -th iteration is formulated as:

$$\mathcal{L} = \mathcal{L}_{align}^I + \mathcal{L}_{align}^O + \lambda_e \mathcal{L}_{align}^{Cross}. \quad (25)$$

Here, the coefficient $\lambda_e = \lambda \cdot \frac{e}{E}$ serves as a dynamic weight that linearly increases from $\frac{\lambda}{E}$ to the maximum value λ as the iteration index e grows from 1 to the maximum iteration E , thereby cautiously and smoothly integrating the complementary knowledge of the two views into the mention representations.

Notably, the matching result between partial cluster hypercubes and type hypercubes gained via the cross-view bridge establishment (Sec. 3.3.1) can be interpreted as an assignment of the matched clusters to their corresponding known types in the ontology. Based on the assignment after the final iteration, the unlabeled entity mentions within these matched clusters can be classified into corresponding known types, while the remaining unmatched clusters (excluding the labeled mentions within them) are regarded as mention clusters of unknown types, yielding the final output result.

4 Experiments

4.1 Experimental Settings

4.1.1 Datasets. To the best of our knowledge, there is no publicly available benchmark dataset for the open-world entity typing (OWET) task. Therefore, we construct two benchmark datasets by adapting existing fine-grained entity typing datasets: BBN [38] and OntoNotes [14]. Details of the construction process are provided in Appendix A. The datasets and source code used in this paper are publicly available for future research¹.

¹https://github.com/CodingPerson/OWET_code

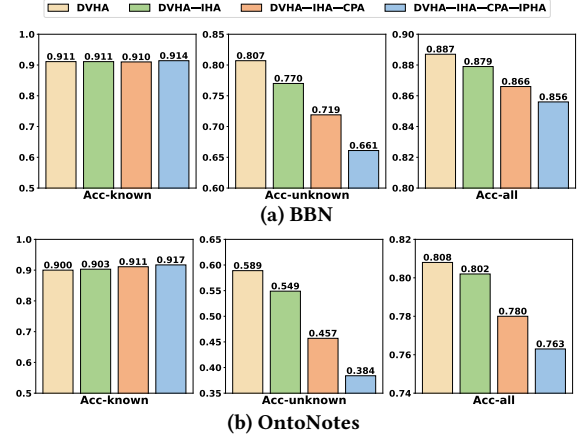


Figure 3: Performance of different variants of DVHA.

4.1.2 Baselines. To our best knowledge, no prior work has been proposed to address the OWET task. For a comprehensive comparison, we adopt eight advanced open-world text classification (OWTC) methods (i.e., CDAC+ [23], DAC [44], MTP [46], USNID [45], DWGF [35], TAN [3], KTN [34] and SDC [2]) and two large language models (LLMs), GPT-4.1 [30] and DeepSeek-R1 [15], as baselines in the OWET task. Detailed descriptions of these baselines are provided in Appendix B, while the setting details and hyperparameter sensitivity analysis of our framework are given in Appendices C and D, respectively.

4.1.3 Evaluation Metrics. Following prior OWTC methods [2, 3, 34], we evaluate the performance of all the methods using overall clustering accuracy (**Acc-all**), known-type clustering accuracy (**Acc-known**) and unknown-type clustering accuracy (**Acc-unknown**). Detailed descriptions of these metrics are provided in Appendix E.

4.2 Performance Comparison

The experimental results of all methods over BBN and OntoNotes datasets are shown in Table 1. From the results, it can be seen that our framework DVHA consistently outperforms all the ten baselines in terms of all metrics by a large margin on both datasets. To be specific, DVHA achieves a substantial performance gain over open-world text classification (OWTC) baselines (i.e., CDAC+, DAC, MTP, USNID, DWGF, TAN, KTN and SDC). This result may be due to the fact that these OWTC baselines inherently disregard the structural knowledge embedded in the type ontology and instead rely solely on the semantic knowledge encoded in mentions, failing to incorporate the reciprocal knowledge, ultimately limiting their

ability to effectively address the OWET task. While LLMs (i.e., GPT-4.1 and DeepSeek-R1) have shown remarkable capabilities across various tasks, they still exhibit inadequacies in tackling the OWET task, which may be attributed to the lack of sufficient prior knowledge tailored to this new task. By contrast, our framework DVHA surpasses all these baselines by leveraging various tailor-designed alignment mechanisms that effectively capture and combine reciprocal knowledge involved in the instance and ontology views, thereby providing a promising way to address the OWET task.

4.3 Effectiveness Study of Different Alignment Mechanisms

To investigate the effectiveness of each alignment mechanism in our framework DVHA, we conduct an extensive evaluation by successively removing different alignment mechanisms to construct the following variants of DVHA: (1) DVHA-IHA, where the intra-view hypercube alignment mechanism (Sec. 3.2.2) is removed; (2) DVHA-IHA-CPA, where both the intra-view hypercube alignment and the cross-view partial alignment (Sec. 3.3) mechanisms are omitted; (3) DVHA-IHA-CPA-IPHA, where the intra-view hypercube alignment, cross-view partial alignment and iterative point-hypercube alignment (Sec. 3.1.2) mechanisms are all excluded. We present the performance of these variants and DVHA in Figure 3.

As these three proposed alignment mechanisms are sequentially removed, we observe a progressive performance degradation in terms of Acc-all over both datasets, clearly validating the effectiveness of each alignment mechanism in our framework. Notably, all variants still maintain competitive and stable performance in Acc-known on both datasets. This result may be attributed to the contribution of mention representation initialization (Sec. 3.1.1) employed in all variants, which sufficiently exploits the known-type supervisory knowledge involved in the labeled set to learn discriminative known-type mention representation. In addition, it can be seen from Figure 3 that these variants exhibit a more pronounced performance decline in terms of Acc-unknown in comparison to their performance in Acc-all. This further demonstrates that these alignment mechanisms enhance the model's ability to discover unknown-type mention clusters, via capturing and combining complementary knowledge embedded in the two views.

4.4 Ablation Study

To verify the superiority of key components in our framework, we conduct a comprehensive ablation study, as reported in Table 2. From the experimental results, we can draw the following observations: (1) For the "w/o AFP" variant, the attentive fusion projection (Sec. 3.1.2) is omitted so that we compute the minimum and maximum corners over all mention representations of a cluster to determine the hypercube boundary. As shown in Table 2, this results in a notable performance decrease in terms of Acc-all on both datasets compared to the whole framework DVHA, which indicates the importance of our attentive fusion projection for effectively projecting each cluster into its corresponding cluster hypercube. (2) Moreover, we construct "w/o GC" and "w/o PC" variants by individually removing the geometric constraint and the probabilistic constraint from DVHA, respectively. It can be seen from the experimental results that both variants suffer from varying degrees of performance degradation on both datasets compared to DVHA,

Table 2: Experimental results of ablation study.

BBN			
Method	Acc-known	Acc-unknown	Acc-all
DVHA	0.911	0.807	0.887
- w/o AFP	0.911	0.763	0.877↓1.0%
- w/o GC	0.910	0.718	0.867↓2.0%
- w/o PC	0.908	0.805	0.884↓0.3%
- w/o DES	0.910	0.746	0.873↓1.4%
- w/o TRANS	0.911	0.784	0.882↓0.5%
- w/o DW	0.911	0.682	0.859↓2.8%
OntoNotes			
Method	Acc-known	Acc-unknown	Acc-all
DVHA	0.900	0.589	0.808
- w/o AFP	0.910	0.531	0.800↓0.8%
- w/o GC	0.910	0.457	0.780↓2.8%
- w/o PC	0.913	0.552	0.808↓0.0%
- w/o DES	0.911	0.551	0.807↓0.1%
- w/o TRANS	0.918	0.532	0.807↓0.1%
- w/o DW	0.910	0.523	0.799↓0.9%

thus verifying the superiority of unifying geometric and probabilistic constraints. Although the "w/o PC" variant achieves comparable performance in Acc-all on the OntoNotes dataset, it exhibits a more pronounced performance decline in Acc-unknown, confirming the benefit of the probabilistic constraint. (3) In the "w/o DES" variant, type descriptions (Sec. 3.2.1) are eliminated and only type names are utilized for the type hypercube projection. This variant results in performance drops on both datasets, which confirms that our generated type descriptions effectively provide comprehensive and distinguishable semantic contexts for type hypercube projection. (4) Additionally, w.r.t DVHA, a performance decline is observed when the cross-view transformation function (Sec. 3.3.2) is omitted in the "w/o TRANS" variant, highlighting its efficacy in cross-view hypercube alignment. (5) Besides, for the "w/o DW" variant, we observe a prominently decreased performance compared to DVHA when the dynamic weight λ_e (Eq. 25) is replaced with a fixed weight λ , which supports our motivation that the dynamic weight is crucial for cautiously and smoothly integrating complementary knowledge from the two views to learn high-quality mention representations.

5 Conclusion

In this paper, we investigate a novel open-world entity typing task (OWET), which seeks to simultaneously classify entity mentions within the known type ontology and discover mention clusters of unknown types. To address this task, we propose a Dual-View Hypercube Alignment (DVHA) framework, which utilizes varied tailor-made alignment mechanisms based on hypercubes to jointly capture and combine reciprocal knowledge involved in the ontology view and the instance view. To effectively integrate the complementary knowledge embedded in both views for better representation modeling, we devise a cross-view partial alignment mechanism by firstly establishing a cross-view bridge and then performing cross-view hypercube alignment between the two views. Experiments over two created OWET benchmark datasets demonstrate that our framework consistently outperforms all baselines.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 62272247, CCF-Huawei Populus Grove Fund, and the Fundamental Research Funds for the Central Universities under Grants 070-63253228.

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You are an AI assistant specializing in entity typing and taxonomic modeling. Your task is to generate semantically precise and ontologically grounded descriptions for entity types in a hierarchical taxonomy. The taxonomy uses slashes (/) to denote parent-child relationships. For example:
 Level 1: Person, Organization
 Level 2: Person/Artist, Person/Athlete, Organization/Company, Organization/Hotel
 For each entity type, write a structured paragraph (3–4 sentences) that:
 • Defines the core semantic scope of the type in relation to its parent;
 • Highlights its defining features that distinguish from other sibling types at the same level, without making explicit comparisons;
 • Optionally includes several real-world entity examples to illustrate the scope.

Figure 4: Prompt for type description generation.

A Dataset Construction Details

We construct two benchmark datasets of the OWET task based on existing FET datasets (i.e., BBN [38] and OntoNotes [14]).

- BBN [38] is extracted from Wall Street Journal articles and annotated with 2 hierarchical levels of 45 types in Freebase. According to the OWET task definition (Sec. 2), we partition the types into known and unknown types, and divide all mentions into a labeled training set, a validation set and an unlabeled (test) set. To provide a comprehensive performance evaluation, it is essential

to incorporate diverse and challenging unknown types with complex topologies. To this end, we designate 11 types as unknown types and regard the remaining 34 types as known (the specific type allocation result is shown in Table 4), where unknown types cover two different categories of topologies:

- (1) a type whose parent type belongs to known types,
- (2) a type whose parent type belongs to unknown types.

Furthermore, the unlabeled set contains both known-type and unknown-type mentions, whereas the labeled and validation sets contain only known-type mentions. On this basis, we construct the unlabeled set by sampling 50% known-type mentions together with all unknown-type mentions, while the remaining known-type mentions are allocated to form the labeled training and validation sets.

- **OntoNotes** [14] is acquired from the newswire documents of the OntoNotes corpus [39] and annotated with 3 hierarchical levels of 57 types. Similar to BBN, we adopt the same manner to divide the types into 43 known types and 14 unknown types (the specific type allocation result is also shown in Table 4), and construct its labeled training, validation, and unlabeled sets.

The statistics of these two constructed datasets are shown in Table 3.

Table 3: Dataset statistics.

Dataset	BBN	OntoNotes
# Known types	34	43
# Unknown types	11	14
# Labeled training mentions	4641	3505
# Validation mentions	1154	863
# Unlabeled known-type mentions	5775	4348
# Unlabeled unknown-type mentions	1712	1758

B Descriptions of Baselines

Open-World Text Classification. Open-world text classification (OWTC), also known as generalized category discovery, is a crucial real-world task that seeks to recognize both known and novel categories from an unlabeled text set by leveraging a labeled text set with only known categories. As the OWET task can be regarded as a variant of OWTC to some extent, we apply eight advanced OWTC methods to the OWET task for comparison.

- CDAC+ [23] is an end-to-end clustering approach that exploits labeled mentions to guide the clustering process, thus discovering mention clusters of unknown types. It further refines the clustering results by encouraging the model to learn from high-confidence cluster assignments.
- DAC [44] effectively transfers prior knowledge from labeled known-type mentions and estimates the number of types by discarding low-confidence clusters. Moreover, it provides stable and reliable supervision signals to guide the clustering process.
- MTP [46] proposes a multi-task pre-training strategy to leverage rich unlabeled mentions along with labeled mentions for representation learning. Furthermore, it designs a new contrastive loss to exploit self-supervisory signals in the unlabeled set for clustering.

- USNID [45] initially captures elementary semantic features through a pre-training stage, and subsequently employs a centroid-guided clustering mechanism to mitigate the inconsistency of cluster assignments during clustering. Furthermore, it discovers fine-grained clusters by jointly optimizing cluster-level and instance-level objectives.
- DWGF [35] is a diffusion-weighted graph framework that captures both semantic similarities and structural relationships inherent in the data. During inference, it further applies a graph smoothing filter to exclude high-frequency noise in ambiguous mentions near cluster boundaries.
- TAN [3] is a transfer and alignment network that integrates two knowledge transfer mechanisms to calibrate the biased knowledge and two feature alignment mechanisms to learn discriminative representations.
- KTN [34] leverages the prediction entropy of a pre-trained classifier to identify unknown-type mentions and determine prototype-based transfer weights to quantify semantic similarities between types, thereby learning discriminative representations and boosting the performance of discovering mention clusters of unknown types.
- SDC [2] dynamically leverages the outputs of a biased pre-trained model to calibrate its own predictions. This procedure mitigates bias toward known types and produces more accurate pseudo-labels for unknown types, thereby enhancing overall performance.

Since these OWTC baselines inherently produce clusters as their final output results, it is challenging to directly assess their classification performance on known-type mentions in the unlabeled (test) set. To ensure a consistent comparison, we adopt the standard evaluation metrics commonly used in OWTC baselines to evaluate the performance of these OWTC baselines as well as our framework DVHA with clustering accuracy on both known-type and unknown-type mentions, which is elaborated in Appendix E.

You are an AI assistant who specializes entity types. Your task is as follows: according to the sentence, predict the entity type of entity mention in the sentence. If the predicted type belongs to the known types, supported by the system, return the corresponding known type; otherwise, suggest the most likely unknown type name (couldn't be 'unknown type', you should give an actual type name different from known types). The supported known types include: **{known types}**. Only provide one type from the above known types, or suggest an unknown type name, and do not give the explanation.
The input format is as follows:
Sentence: <sentence> Mention: <mention>

Figure 5: Prompt for LLM baselines.

Large Language Models. To apply LLMs to the OWET task, we provide a tailor-designed prompt that instructs the LLM to predict an appropriate type for each entity mention, as illustrated in Figure 5. If the predicted type belongs to known types, the LLM outputs the corresponding known type; otherwise, it suggests the most likely unknown type name.

- GPT-4.1 [30] is an advanced large language model that combines strong reasoning ability with high efficiency and responsiveness, enabling accurate, scalable, and versatile natural language understanding and generation. In our experiments, we adopt

Table 4: Entity Type Composition for Datasets.

BBN	OntoNotes
Known Types in BBN: /person, /organization, /location, /gpe, /product, /facilities, /disease, /event, /game, /plant, /language, /contact_info, /substance, /organization/corporation, /organization/government, /organization/hotel, /organization/hospital, /organization/political, /organization/religious, /location/continent, /location/lake_sea_ocean, /location/region, /gpe/city, /gpe/state_province, /product/vehicle, /product/weapon, /facility/airport, /facility/attraction, /facility/bridge, /facility/building, /facility/highway_street, /event/hurricane, /substance/chemical, /substance/drug	Known Types in OntoNotes: /person, /organization, /location, /other, /person/political_figure, /person/legal, /organization/government, /organization/stock_exchange, /organization/political_party, /organization/sports_team, /organization/education, /organization/military, /location/country, /location/city, /location/structure, /location/geography, /location/park, /location/transit, /other/currency, /other/health, /other/event, /other/product, /other/living_thing, /other/sports_and_leisure, /other/body_part, /other/heritage, /other/scientific, /other/supernatural, /location/structure/government, /location/structure/hotel, /location/structure/sports_facility, /location/transit/railway, /location/transit/road, /other/health/treatment, /other/event/sports_event, /other/event/protest, /other/event/holiday, /other/event/natural_disaster, /other/event/election, /other/product/weapon, /other/product/computer, /other/product/software, /other/living_thing/animal
Unknown Types in BBN: /work_of_art, /law, /animal, /gpe/country, /organization/educational, /event/war, /substance/food, /location/river, /work_of_art/book, /work_of_art/play, /work_of_art/song	Unknown Types in OntoNotes: /person/artist, /person/title, /organization/company, /other/art, /other/legal, /other/food, /person/artist/author, /organization/company/broadcast, /organization/company/news, /location/geography/body_of_water, /other/art/broadcast, /other/art/writing, /other/product/car, /other/event/violent_conflict

GPT-4.1-nano, a lightweight and cost-efficient variant of GPT-4.1, as a baseline and evaluate its performance on the OWET task with the above prompt.

- DeepSeek-R1 [15] is a reasoning-oriented large language model trained via large-scale reinforcement learning, demonstrating remarkable reasoning capabilities. We evaluate its performance for the OWET task in the same way as GPT-4.1.

It is worth noting that the predicted unknown types output by LLM baselines may not consistently align with ground-truth labels of unknown types, making evaluation on unknown-type mentions infeasible. To address this issue, we convert the predicted results into clustering results according to the predicted type labels and adopt the same evaluation manner as our framework DVHA and OWTC baselines, thereby enabling a fair and consistent comparison of LLM performance.

C Setting Details

In our DVHA framework, we select BERT [9] as the default PLM (Sec. 3.1.1), and GPT-4.1 [30] for type description generation (Sec. 3.2.1). The dimension d for both mention representations and hypercubes is set to 128. The margin hyperparameter γ (Eq. 6) is set to 35 and 10 over BBN and OntoNotes, respectively. The temperature parameter τ (Eq. 7) is fixed at 16. The positive geometric margin δ (Eq. 12) and the negative geometric margin ϵ (Eq. 14) are set to 0.01. The minimum offset size ϕ^I (Eq. 8) and ϕ^O (Eq. 18) are set to 0.1 and 0.5, respectively. The control parameter α (Eq. 19) is set to 0.1.

The maximum value λ (Eq. 25) is set to 3 and the maximum iteration E is set to 100. For optimization, we use Adam with a learning rate of $2e-5$ for BERT and $1e-4$ for other modules. All experiments are implemented using PyTorch with one NVIDIA RTX A6000 GPU.

D Hyperparameter Sensitivity Analysis

To assess the robustness of our framework, we conduct a comprehensive sensitivity analysis on both datasets with respect to five key hyperparameters: the positive geometric margin δ (Eq. 12), the negative geometric margin ϵ (Eq. 14), the minimum offset size ϕ^I (Eq. 8) of cluster hypercubes, the minimum offset size ϕ^O (Eq. 18) of type hypercubes and the control parameter α (Eq. 19).

Regarding the positive geometric margin δ and negative geometric margin ϵ , as depicted in Figure 6 (a) and 6 (b), our framework exhibits stable performance in terms of Acc-all across a range of δ (0 to 0.02) and ϵ (0 to 0.02) on both datasets, indicating our framework’s robustness and insensitivity to δ and ϵ . For the minimum offset size ϕ^I , as depicted in Figure 6 (c), the performance remains stable across a range of ϕ^I from 0 to 0.2 on OntoNotes. In contrast, on BBN, the performance decreases when $\phi^I < 0.1$, but remains competitive and stable when $\phi^I \geq 0.1$, indicating that the framework exhibits robustness when ϕ^I is not excessively small. Additionally, it can be seen from Figure 6 (d) and 6 (e) that our framework DVHA maintains stable performance across a broad

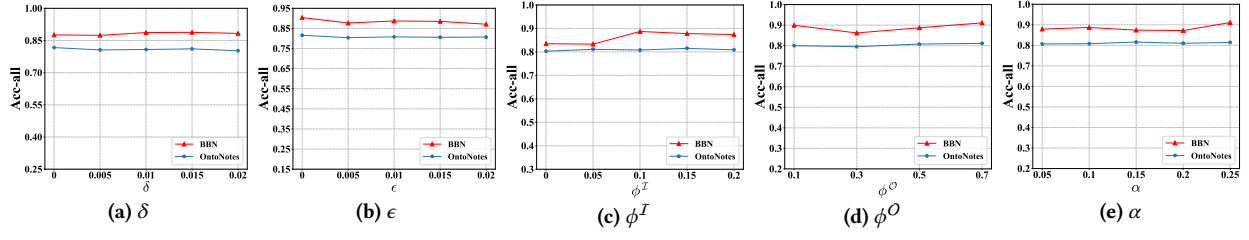


Figure 6: Hyperparameter sensitivity analysis.

range of ϕ^O (0.1 to 0.7) and α (0.05 to 0.25) on both datasets, confirming the framework’s stability with respect to the minimum offset size ϕ^O and the control parameter α .

E Evaluation Metrics

In order to conduct a fair and consistent comparison with the aforementioned baselines, we follow the standard evaluation metrics adopted in OWTC baselines [2, 3, 34] to evaluate the performance of all the methods with clustering accuracy on the unlabeled set $\mathcal{M}_u$. Specifically, we employ the Hungarian algorithm [21] to align generated clusters with type labels for the unlabeled set $\mathcal{M}_u$ and then leverage the aligned type labels of clusters to measure the clustering accuracy. Our primary evaluation metric is the overall clustering accuracy (**Acc-all**) for all mentions within the unlabeled set $\mathcal{M}_u$. To provide a comprehensive performance evaluation, we further calculate the known-type clustering accuracy (**Acc-known**) for known-type mentions in $\mathcal{M}_u$, as well as unknown-type clustering accuracy (**Acc-unknown**) for unknown-type mentions in $\mathcal{M}_u$.

F Related Work

F.1 Fine-Grained Entity Typing

Fine-grained entity typing (FET) seeks to assign each entity mention an appropriate fine-grained type given its surrounding context and a predefined type ontology. A major line of FET methods emphasizes modeling the hierarchical structure of the ontology [19, 25, 28, 41, 47]. For example, HMGCN [19] introduces a hierarchical multi-graph convolutional network to capture diverse semantic relatedness between entities. Moreover, to enhance the discriminability between closely related types, PICOT [47] proposes a contrastive learning strategy that explicitly models type distinctions. Another line of research reformulates the FET task into other tasks that exploit more effective label semantics [10, 12, 16]. Concretely, PLET [10] designs entity-oriented prompts and formalizes FET as a cloze-style task, while CASENT [12] redefines FET as a sequence-to-sequence generation problem, thereby predicting entity types with calibrated confidence scores. Nevertheless, these FET methods adhere to the closed-world assumption, failing to discover mention clusters of unknown types, and thus cannot be applied to the OWET task.

F.2 Open-World Text Classification

Open-world text classification (OWTC), a.k.a. generalized category discovery, aims to recognize both known and novel categories

from an unlabeled text dataset by leveraging a labeled text dataset with only known categories. Owing to the absence of supervision signals for novel categories, OWTC methods primarily rely on pseudo-labeling strategies [23, 44–46] to learn from the unlabeled set. For example, CDAC+ [23] employs pseudo supervision based on pairwise similarities to guide the clustering process, while DAC [44] adopts an alignment strategy during clustering to generate pseudo labels for known and novel categories. Another line of work incorporates transfer learning to improve representation learning [2, 3, 34]. For instance, TAN [3] integrates two knowledge transfer mechanisms to learn discriminative representations and KTN [34] leverages prototype-based transfer weights to measure semantic similarities between categories, thereby transferring knowledge from known categories to novel categories.

However, all aforementioned works exclusively exploit semantic knowledge embodied in mentions, omitting the structural knowledge inherent in the type ontology, resulting in their inability to effectively address the OWET task, as shown in our experiments.

F.3 Representation Learning with Hypercube

Hypercubes have emerged as a powerful geometric representation approach, supporting the remarkable success for a wide range of applications. For example, Query2Box [33] encodes queries and entities as hypercubes in the knowledge graph, enabling diverse logical reasoning. CubeRec [7] designs a hypercube recommender system by representing groups of users as hypercubes for group recommendation. For digital education, BoxCD [13] utilizes hypercubes to represent learners and exercises, with the volume of intersecting hypercubes used to predict learners’ responses. Moreover, BoxRetrieval [26] advances ranking tasks by parameterizing both queries and items as hypercubes to enhance effectiveness and efficiency. However, to the best of our knowledge, our paper is the first work to investigate the open-world entity typing task with hypercubes.